Dhruv Shah

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Applied mathematician and control theorist moving into generative modeling. My Ph.D. centers on Koopman operator theory, which finds accurate linear representations of nonlinear dynamics, and I have shown that this machinery provides a principled, structured architecture for few-step flow maps and consistency models. My research advances fundamental questions in Koopman theory and turns those advances into a new angle on generative AI. I am looking to bring this expertise, along with a broad background in the engineering sciences, to problems in generative AI and robotics.

EDUCATION

University of California, San Diego — Sep 2023 – Present
Ph.D. (from Jul 2024) and M.S. (Jun 2024), Mechanical and Aerospace Engineering — Advisor: Prof. Jorge Cortés

Indian Institute of Technology Bombay — Aug 2019 – May 2023
B.Tech. in Electrical Engineering with Honors, Minor in Computer Science and Engineering — Advisor: Prof. Vivek Borkar

SKILLS

Generative modeling	Flow matching, flow maps / consistency models, Diffusion Transformers, optimal-transport couplings
Mathematics	dynamical systems, convex optimization, probability theory, kernel methods, numerical linear algebra
Control theory	Robust & nonlinear MPC, optimal control, model reduction, system ID, Kalman & particle filtering
Robotics	3D rotations & Lie groups, Robot Kinematics, Sensor fusion, SLAM, VLAs, World Models
Tools	Python, PyTorch, JAX, multi-GPU training, SLURM, Kubernetes, MATLAB, CasADi, CVXPY

FEATURED RESEARCH: KOOPMAN FLOW MAPS FOR FEW-STEP GENERATION

Koopman Flow Maps: Mode Structure for Flow Map Matching *(in preparation)* — UCSD, 2025 – Present

- Proposed **Koopman Flow Maps (KFMs)**, a structured flow-map parametrization inspired by Koopman theory: a DiT backbone sees only the departure time, and the jump to any arrival time is a closed-form function of a learned spectrum.
- **Theory:** proved that the KFM structure incurs no loss of expressivity, and gave a novel interpretation of the existing self-distillation losses in the flow-map matching literature through the Koopman lens.
- **Results:** with matched backbone, loss and training budget, the KFM-parametrized network beats its unstructured counterpart at *every* NFE from 1 to 64 on both CIFAR-10 and CelebA-HQ, trained from scratch.
- **Engineering:** implemented DiT flow-map training (iMF, LSD, ESD, PSD and Koopman variants), CFG, and FID pipelines from scratch; ran multi-GPU A100 training on a Kubernetes cluster (NRP Nautilus).

RESEARCH FOUNDATIONS AND RELEVANT BACKGROUND

Koopman operators: linear models of nonlinear dynamics — UCSD, 2024 – Present

- **Subspace pruning:** prune a large candidate dictionary to the subspace best suited to the dynamics. *(CDC 2026; TAC, submitted)* Unified existing pruning methods, fixed numerical drift (**4 orders of magnitude** more accurate), **14**× faster using rank-one updates
- **Kernel pruning:** extended subspace pruning to kernel-based features, which are flexible but expensive at scale. *(CDC 2026)* Used Nyström features to cut the cost from cubic in dataset size to near-linear while matching the exact method.
- **Discontinuous Koopman:** systems with several long-term behaviors defeat a single lifted linear model with continuous liftings. Learned to split the state space by long-term behavior, one model per region: **15–26**× lower long-horizon error. *(ACC 2027)*

Control theory and engineering — UCSD, 2023 – 2024

- **Neural feedback loops:** LMI stability certificates via LFR + incremental QCs; LPV controller synthesis. *(L-CSS 2025)*
- **System identification:** Bose ANC headset from real data: spectral analysis, Hankel realization, 4th-order Box–Jenkins model.
- **State estimation:** Kalman filters for correlated and colored noise; MCMC particle-filter resampler, **3**× faster.
- **Robust nonlinear MPC** (UCSB MPC Workshop, Rawlings & Diehl): Theory and hands on excercises using CasADi, acados.

Robotics: sensing and state estimation *(ECE 276A projects)* — UCSD, 2026

- **IMU orientation tracking:** recalibrated the IMU against VICON; recast quaternion estimation as a shooting problem over angular velocities (PyTorch), cutting RMS error on all 7 datasets (up to **63%**); built camera panoramas.
- **LiDAR SLAM:** encoder/IMU odometry warm-starts ICP scan matching; fixed-interval and proximity loop closures in a GTSAM pose graph remove long-term drift, yielding a globally consistent occupancy grid map.
- **Visual-inertial SLAM:** $SE(3)$ EKF fusing IMU and stereo camera on real robot data; banded sparse covariance to track **1,500+** landmarks, stereo outlier rejection, and noise tuning via innovation whiteness.

SERVICE AND AWARDS

- **Session co-chair:** "Advances in the Koopman Operator for Systems and Control," CDC 2026. **Reviewer:** TAC, L-CSS, CDC, ACC.
- **Awards:** IIT Bombay Undergraduate Research Award (2022), MITACS Globalink Fellowship (2022), DAAD-WISE Scholarship (2022); KVPY Fellowship (2019).